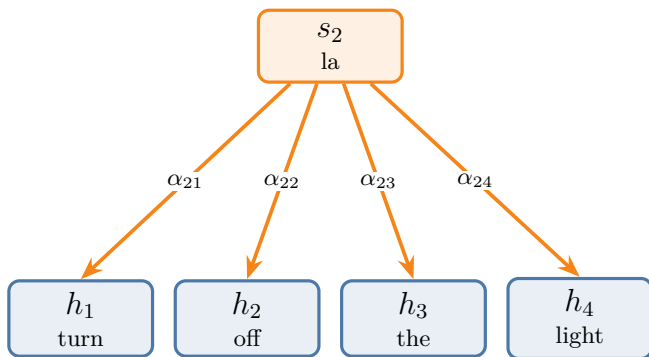


Luong attention: two sequences

decoder state: the query (French)

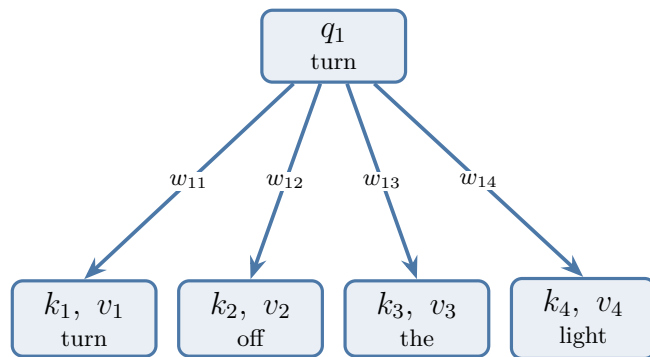


encoder states: keys and values (English)

$$c_2 = \alpha_{21}h_1 + \alpha_{22}h_2 + \alpha_{23}h_3 + \alpha_{24}h_4$$

Self-attention: one sequence

the query (the same English sentence)



keys and values (English)

$$y_1 = w_{11}v_1 + w_{12}v_2 + w_{13}v_3 + w_{14}v_4$$

	Luong attention	self-attention
query	decoder state s_i	q_i of a word in the sentence
key	encoder state h_j	k_j of a word in the same sentence
value	encoder state h_j	v_j of a word in the same sentence
weights	3×4 : French $\times$ English	4×4 : English $\times$ English